

LESLIE KADENGE

Software Engineer

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SUMMARY

Information Technology graduate working through a public portfolio spanning data science, backend services, front-end accessibility and retrieval. Eleven repositories, each with reproducible code, tests, and a README stating what was measured — including the results that were negative.

The consistent thread is the bug that leaves no trace. An authorisation check that never runs. A query plan that is quietly quadratic. A test suite reporting success while watching nothing.

SELECTED PROJECTS

Job Board FastAPI · React · SQLite · Docker

github.com/les-k/job-board

bullet 224 tests, 54 of them an access-control matrix proving the wrong person cannot act. Verified by deleting the ownership check: exactly three tests failed and everything else passed.

bullet A full-text search that ran a match once per listing — 145 ms against 0.02 ms once rewritten. Keyset pagination with a test reproducing the row OFFSET duplicates.

Credit Card Fraud Detection Python · scikit-learn · FastAPI

github.com/les-k/fraud-detection

bullet 0.17% positive rate, where flagging nothing scores 99.87% accuracy. Threshold set from an explicit cost model, recovering \$12,168 across a 48-hour holdout at 82% recall.

Semantic Search Python · sentence-transformers

github.com/les-k/semantic-search

bullet Six retrieval methods over 1,400 documents and 1,837 relevance judgments. Two of five claimed improvements failed a paired significance test, including one better on every average.

snip — deployed URL shortener FastAPI · Docker · live

github.com/les-k/snip

bullet 34 tests at 95% coverage, CI across three operating systems and five Python versions, refusing redirects to private and cloud-metadata addresses.

Component Library React 19 · WCAG AA

github.com/les-k/component-library

bullet Nine accessible components where 34 contrast pairings and 18 structural guards gate the build, so an inaccessible change cannot merge.

Rafiki — voice assistant in Kiswahili Python · Chrome extension

github.com/les-k/Rafiki-Digital-assistant-AI

bullet A voice assistant whose command vocabulary is entirely Kiswahili, a language none of Alexa, Siri or Google Assistant support.

sweep — command-line tool Python · zero dependencies

github.com/les-k/sweep

bullet Deletes a directory only when a marker file proves it is regenerable. 75 tests, CI on three operating systems.

ALSO PUBLISHED

wine-quality — duplicate rows leaking across a train/test split, inflating accuracy by 11 points. · customer-segmentation — the same analysis in pandas and SQL, cross-checked to four decimals, reporting a negative clustering result as one. · sales-prediction — newspaper spend indistinguishable from noise at $p = 0.72$. · car-price-prediction — a rule of thumb reaching $R^2 0.555$ before any model, set as the bar to beat. · 2048 — game rules separated from rendering and covered by 75 tests. · sweep-landing — Lighthouse 100 across all four categories at 22.6 KB, with a contrast check gating CI.

TECHNICAL SKILLS

LANGUAGES Python, JavaScript, SQL, HTML, CSS

BACKEND FastAPI, SQLite, REST API design, authentication, access control

FRONTEND React 19, Vite, accessible components, WCAG AA

DATA & ML pandas, NumPy, SciPy, scikit-learn, sentence-transformers, DuckDB

TESTING pytest, node:test, coverage, mutation testing, property-based testing

OPERATIONS Docker, GitHub Actions, Render, CI matrices across OS and runtime

ACADEMIC PROJECT

MOODS — a final-year sentiment analysis model classifying posts as positive, negative or neutral.

EDUCATION

BSc Information Technology — Zetech University, November 2023 *Second-class honours, upper division*

Coursework: Machine Learning, Data Science, Artificial Intelligence, Database Systems, Software Engineering, Distributed Systems, Programming Fundamentals (Python, Java, C, C++, JavaScript).

CERTIFICATION

ICDL Profile — Computer Pride.

References available on request.